CAP Theorem - Consistency

1. What is CAP Theorem?

CAP theorem states that a distributed system can only strongly guarantee **two out of three** properties at the same time:

```None
C = Consistency

A = Availability

P = Partition Tolerance
```

During a network partition, you usually choose between **Consistency** or **Availability**.

2. Consistency in CAP (Important)

CAP consistency means:

```None
Every read gets the latest successful write
```

All nodes behave as if there is a **single up-to-date copy** of data.

This is closer to **Linearizability**, not the same as ACID consistency.

3. Simple Meaning

If user updates balance to:

None

```
₹5000
```

Then immediately after write:

- Node A returns ₹5000
- Node B returns ₹5000
- Node C returns ₹5000

All users see same latest value.

4. Example

Distributed user profile system:

None

```
Name changed to "Samarth"
```

Any node queried after successful write should return:

None

```
Samarth
```

No stale value allowed.

5. Real-World Analogy

ATM network:

If you withdraw money from one ATM, another ATM should immediately know updated balance.

That is consistency expectation.

6. Why It Matters

Consistency is important in:

Banking
Payments
Inventory count
Order confirmation
Seat booking
Identity/auth systems

7. How It Is Achieved

Systems may use:

- Synchronous replication
- Leader-based writes
- Consensus protocols
- Quorum reads/writes

Examples:

- ZooKeeper
- etcd
- Google Spanner

8. Cost of Strong Consistency

To ensure same latest data everywhere:

Higher latency
Lower availability during partitions
More coordination overhead
Slower writes sometimes

9. Partition Example

Suppose Node A cannot talk to Node B.

If user writes new data to A:

System must choose:

```None
Option 1: Reject request → keep consistency

Option 2: Accept request → lose consistency temporarily
```

This is core CAP tradeoff.

10. Consistency vs Eventual Consistency

Type	Meaning
Strong Consistency	Latest write always visible
Eventual Consistency	Nodes become same later

Example

After changing profile photo:

- Strong consistency → all nodes instantly updated
- Eventual consistency → some users may see old photo briefly

11. CAP Consistency ≠ ACID Consistency

Very important interview point.

CAP Consistency	ACID Consistency
Same latest data on nodes	Valid database state / constraints

Distributed systems concept	Transaction concept

12. HLD Design Use Cases

Choose strong consistency for:

```None
Wallet balances

Payment ledger

Ticket inventory

Stock quantity
```

Choose weaker consistency for:

```None
Likes count

View count

Analytics

Recommendations

Feed ranking
```

13. Interview Answer Example

Q: What is consistency in CAP theorem?

> Consistency means every read receives the most recent successful write, so all nodes return the same latest data despite replication.

14. Short Revision Notes

```None
CAP Consistency:

All reads see latest write



Same data on all nodes



Usually costs:

- higher latency

- lower availability during partition
```

15. One-Line Summary

Consistency in CAP means every node returns the same most recent data, giving users a single correct view of distributed state.